

UNIT 1 – GENETICS

Contrasting characters of *Pisum sativum* considered by Mendel

Dominant and recessive characters

Monohybrid cross

Law of Dominance with an example

Law of Segregation with an example

Dihybrid cross

Law of Independent assortment with an example

Test cross and Back Cross

Mendel's Laws of Inheritance - Monohybrid cross

Genetics is the science that explains the mechanism of Inheritance of characters from the parent to the offspring.

Individuals of any species are generally like the parents that produced them or children tend to resemble their parents. No doubt children resemble their parents, yet it is seen that they are not their exact duplicates. Thus here we encounter two entirely different phenomena that are called heredity and variation. Heredity is the transmission of characters from one generation to successive generations whereas differences among the living things are called variations. The way in which characters are transmitted from one generation to another was first demonstrated by “Gregor Johann Mendel” in 1866 and he is considered as “Father of genetics” Mendel was not the first to conduct hybridization experiments, but he was the first to consider one trait at one time and this was perhaps the secret of his success.

REASONS FOR THE SELECTION OF PEA PLANT BY MENDEL

Mendel chose garden pea (*Pisum sativum*) for his experiments. The pea plant shows a number of clear cut contrasting characters.

- 1) Peas were obtainable in many pure breeding varieties.
- 2) The flowers were well protected from the influence of foreign pollen, owing to the close encasement of the reproductive organs inside the petals of the flower.
- 3) The hybrid resulting from crossing two varieties were perfectly fertile
- 4) It has perfect bisexual flowers containing both female and male parts and the flowers are predominantly self-pollinating.
- 5) Easy hybridization.

- 6) It is an annual plant-Its short life cycle made it possible to study several generations within a short period.

He maintained statistical records of the results. It helped Mendel to derive to numerical ratios of significance.

CONTRASTING CHARACTERS OF PEA PLANT STUDIED BY MENDEL

Mendel took into consideration **seven characters** of the garden pea. Each of these seven characters has two varieties (alternatives) which are mentioned here under.

No.	Character	Contrasting form / traits	
		Dominant	Recessive
i.	Height of stem	Tall (TT)	Dwarf (tt)
ii.	Colour of flower	Coloured (CC)	White (cc)
iii.	Position of flower	Axial (AA)	Terminal (aa)
iv.	Pod shape	Inflated (II)	Constricted (ii)
v.	Pod colour	Green (GG)	Yellow (gg)
vi.	Seed shape	Round (RR)	Wrinkled (rr)
vii.	Seed colour (cotyledon)	Yellow (YY)	Green (yy)

Seven Traits of Mendel's Pea Plants		
Trait	Dominant Allele	Recessive Allele
Height	 Tall	 Dwarf
Flower Position	 Axial Flowers located along the stem	 Terminal Flowers located at the stem tips
Flower Color	 Purple	 White
Pod Color	 Green	 Yellow
Pod Shape	 Smooth	 Constricted
Seed Color	 Yellow	 Green
Seed Shape	 Round	 Wrinkled

MENDEL'S EXPERIMENTS

MONOHYBRID CROSS

In his simpler experiments Mendel crossed two plants differing in one character only ex: a plant having tall stem was crossed with a plant having dwarf stem and so on. Such crosses are known as monohybrid crosses (crosses where parents differ in one pair of alternative characters) and the hybrids thus produced are known as monohybrids.

Crossing of genetically different individuals is called hybridization. The plant in which the hybridization is made is known as parental generation which is denoted by 'P'.

The generation of hybrids resulting from a cross between parental plants is called the first filial generation (filial means off springs) and is denoted by F₁. The second generation of hybrids arising from the F₁ generation by self or cross fertilization is called second filial generation and is denoted by F₂.

Mendel selected plants which were pure (A pure plant is one that breeds true for a particular character) Mendel crossed the pure tall plant with the pure dwarf plant. To do this he transferred Pollen grain of tall plant (male parent) to the stigma of emasculated (removing anthers from the flowers of female parent) flower of dwarf plant (female parent) or vice versa. The resultant progeny of F₁ generation were found to be tall. What happened to the recessive allele? To find out this, he continued his experiments further and allowed self pollination among the F₁ plants. In F₂ both tall and dwarf were produced in the ratio of 3:1. This ratio is known as monohybrid ratio.

Explanation: According to Mendel, the expression of each character is controlled by two factors (Mendel's factors are now called as genes). Each factor is received from each parent. The dominant factor is represented by capital letter of the first letter of that character. The recessive factor is represented by small letter of the first letter of the dominant character. The factors for tallness and dwarfness are represented by the letters 'T' and 't' respectively. Hence genotype of homozygous tall plant is 'TT' and that of

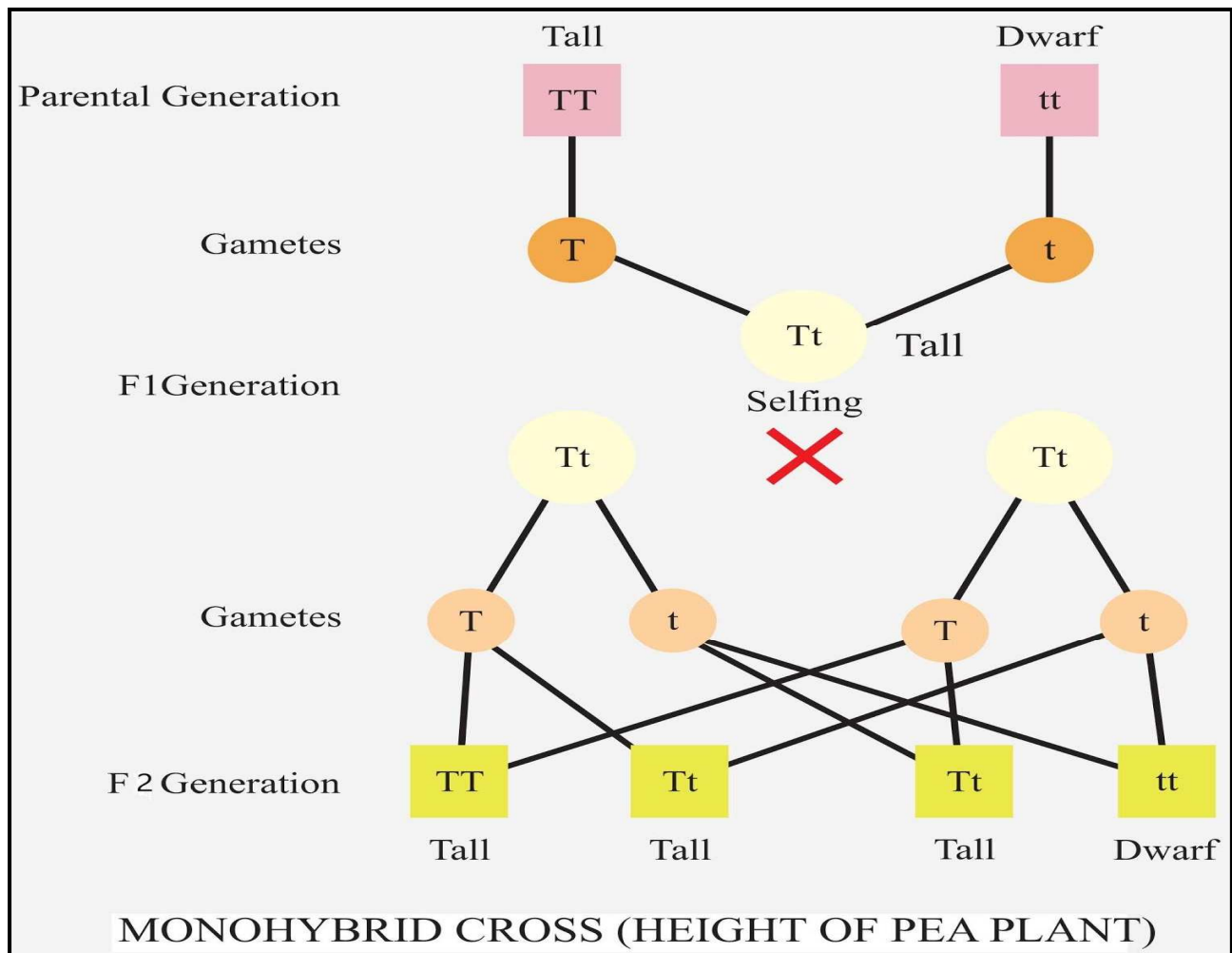
dwarf plant is 'tt' such factors responsible for controlling the same contrasting characters are called as allelomorphs or alleles.

During fertilization in a cross between a tall and dwarf plant, the tall plant contributes gametes, each containing 'T' and the dwarf plant contributes gametes each containing a 't'. After fertilization, the F₁ offsprings will have both these factors, 'Tt'. In appearance, these plants having 'Tt' are tall because 'T' is dominant and 't' is recessive. The progeny obtained in F₁ generation were heterozygous. When F₁ hybrid produces gametes, half of the gametes carry the factor 'T' and remaining half carry the factor 't'. Thus the gametes are pure and each gamete contains only one factor and never both. When F₁ hybrids (Tt) are self-pollinated the two entities separate out and unite independently producing tall and dwarf plants in the ratio 3:1 in F₂ generation. The F₂ generation contain pure or homozygous tall (TT), hybrid tall or heterozygous tall (Tt), and pure dwarf (tt) in the ratio of 1:2:1. It is the genotypic ratio. But the pure tall (TT) and hybrid tall (Tt) have the same phenotype. So the phenotypic ratio in F₂ is 3:1.

The above explanation is shown in the 'Checker Board' or "Punnet squares". All possible combinations or unions of the different male and female gametes during fertilization are represented by this Punnet squares.

F₂ generation phenotypic ratio = 3:1 (3 tall, 1 dwarf)

F₂ generation genotypic ratio = 1:2:1 (1TT, 2Tt, 1tt)



Based on the findings of monohybrid experiments Mendel proposed the following laws:

1. Law of Dominance

This law states that one factor (gene) in a pair may mask or prevent expression of the other. He called the forms that appeared in the F1 generation of his monohybrid cross as dominant and those which did not appear as recessive.

2. Law of Segregation or Law of purity of gametes

This law states that during gamete formation, the two factors (genes) of each character segregate (separate) leaving one factor of a character in each gamete. Thus according to this law a gamete is always pure for a character and never a hybrid. Hence this law is also called as 'Law of purity of gametes'.

DIHYBRID CROSS

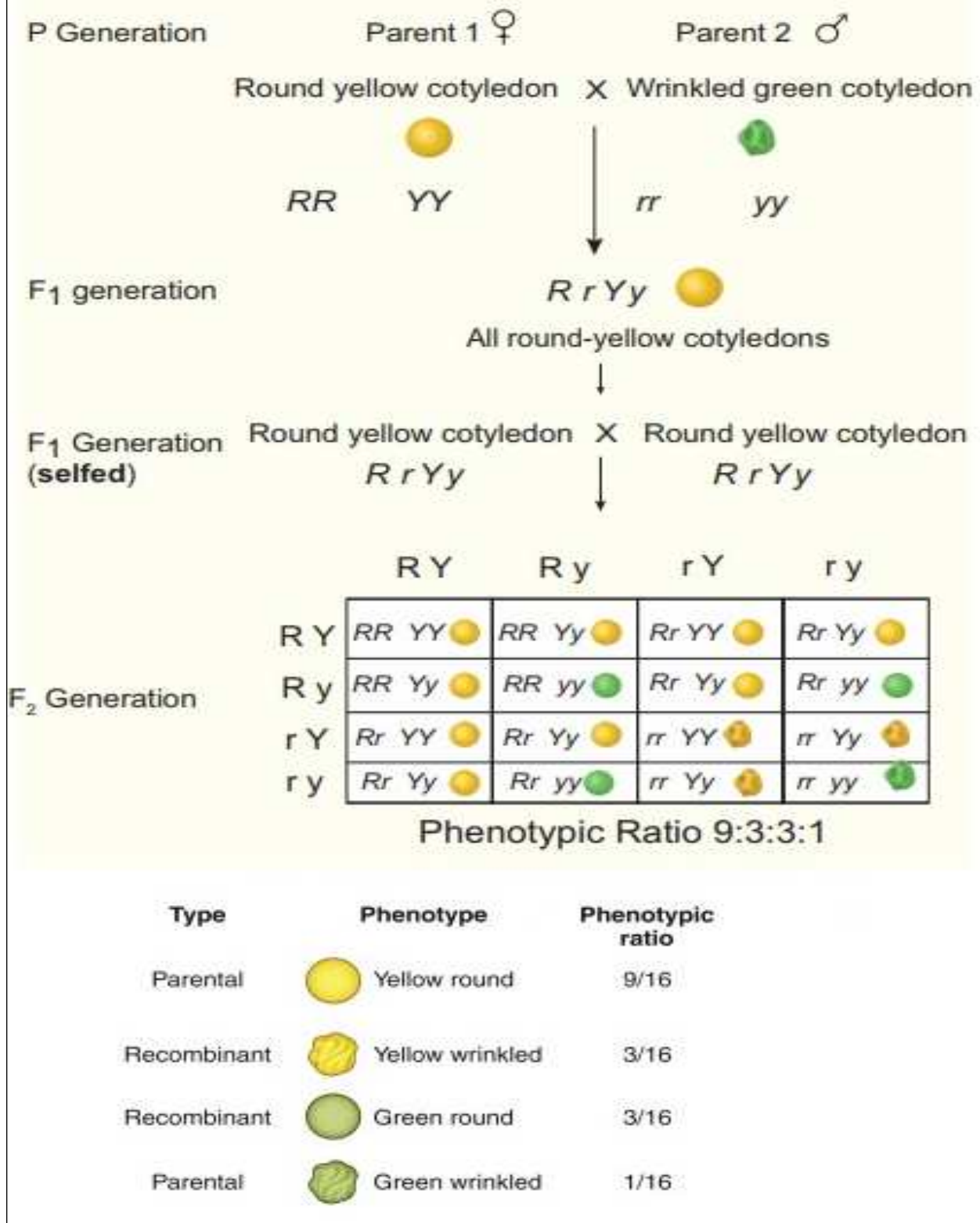
After Monohybrid experiments, Mendel tried to analyse cases where two pairs of characters were simultaneously studied. He tried to find out whether one pair of characters is completely independent of another pair or else if there was any deviation. As we know, crosses where two pairs are involved are known as dihybrid crosses. For Dihybrid crosses Mendel selected pea plants which were pure for two characters. He crossed a plant pure for round and yellow seeds with the plant pure for wrinkled and green seeds. All the F₁ plants showed round and yellow seeds, as these two characters are found to be dominant over the other two characters. The F₁ plants were allowed to produce the F₂ generation by self fertilization.

In the F₂ generation, of the 16 possible combinations in the F₂ generation, four varieties appeared in the following combination

- 1) Round- yellow (both dominant characters) 9
- 2) Round- green (one dominant and one recessive character) 3
- 3) Wrinkled-yellow (one dominant and one recessive) 3
- 4) Wrinkled-green (both recessive) 1

This ratio of **9:3:3:1** is called **Dihybrid ratio**.

Dihybrid cross



Explanation

Mendel represented round character of seed by “R” and wrinkled by “r”, similarly yellow character by “Y” and green by “y”. The genotype of pure dominant round and yellow seeded plant is “RRYY”. The genotype of pure recessive wrinkled and green seeded plant is “rryy”. The gametes produced from “RRYY” are “RY” type and the gametes produced from “rryy” are “ry” type.

When they are crossed, in the F₁ generation all the hybrids contain “RrYy”. All are round and yellow seeded plants. The gametes of these plants contain only one of the two factors of each allelomorphic pair. Thus a gamete may have either ‘R’ or ‘r’ in addition to either ‘Y’ or ‘y’. Four types of gametes are produced in F₁ generation, they are “RY, Ry, rY and ry”. When F₁ dihybrids are self pollinated, in the F₂ generation 16 possible combinations are observed, as shown in the checker board, in the ratio of 9:3:3:1. It is the dihybrid phenotypic ratio. The genotypic ratio of dihybrid cross is 1:2:1:2:4:2:1:2:1. This experiment shows that rounded form does not remain always with yellowness, but it can combine with greenness as well. Similarly wrinkled form can go with yellowness. From these findings Mendel formulated his third law known as Law of Independent assortment.

Law of Independent Assortment

It states that factors (genes) for different pairs of contrasting characters are assorted independent of one another during gamete formation. This implies that during gamete formation, the alleles for colour of seed do not interfere with alleles for shape of seed and the two pairs of alleles behave independent of each other.

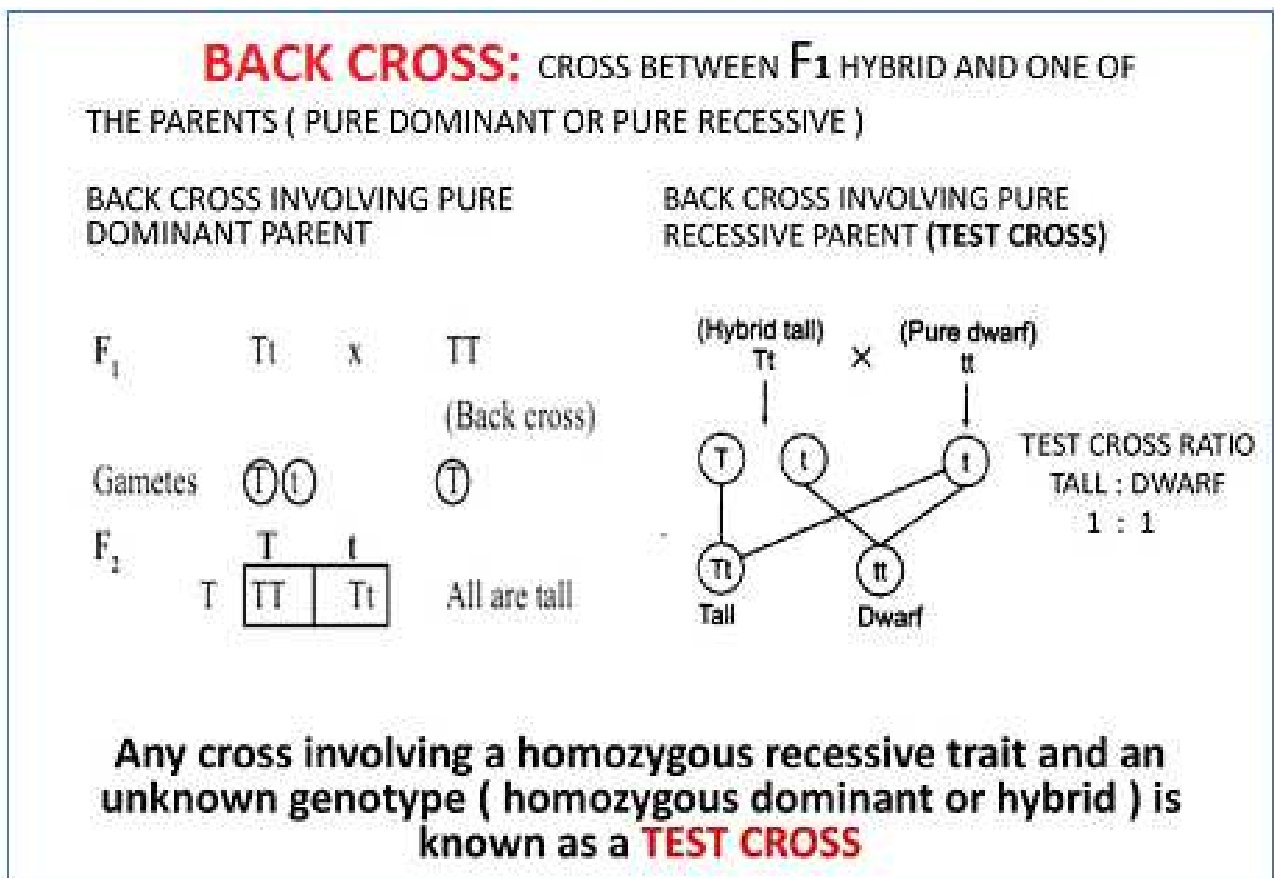
BACKCROSS AND TEST CROSS

Crosses between F₁ off springs with either of the two parents are known as Back crosses.

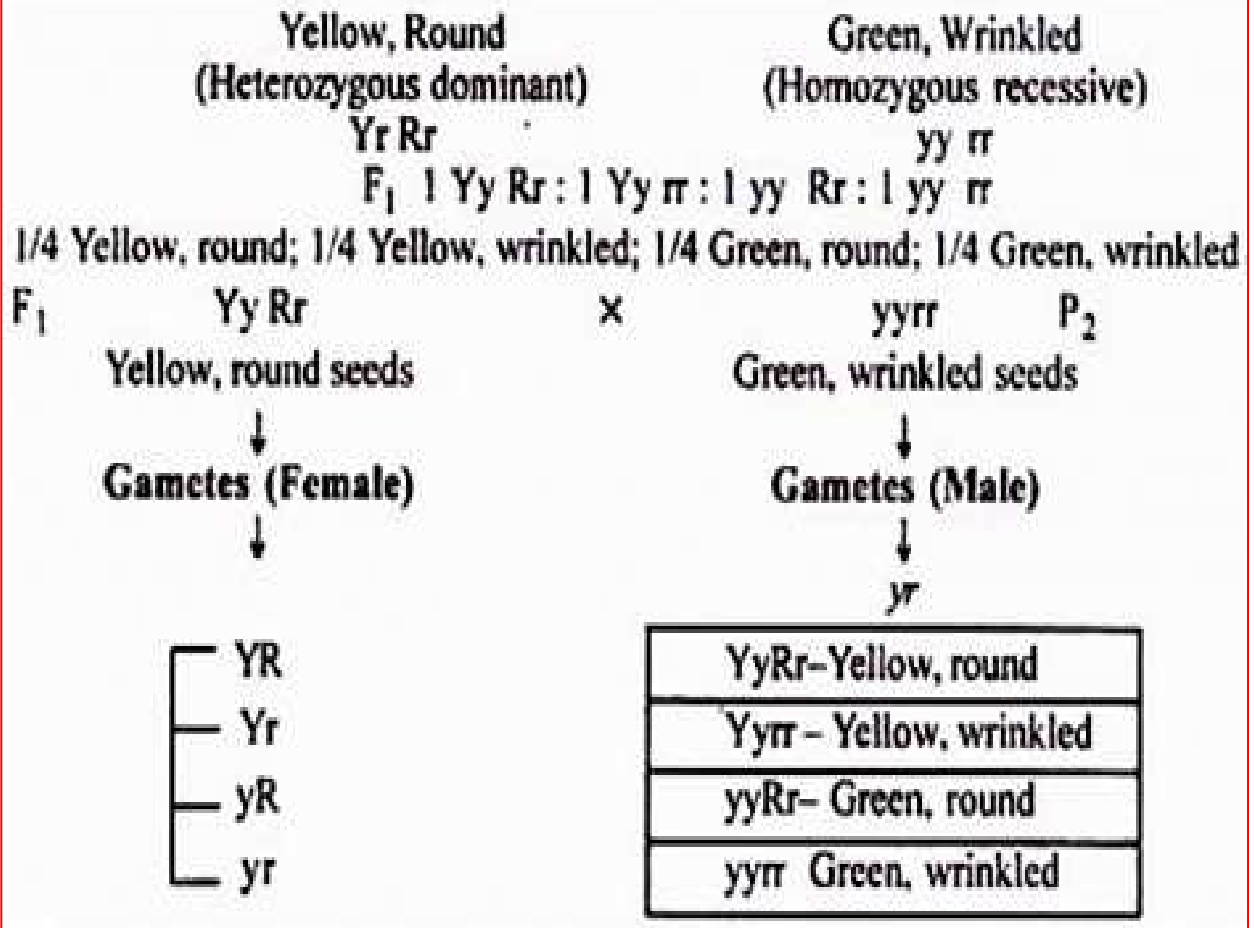
If F₁ off springs are crossed with the **dominant parents** all the F₂ off springs develop **dominant character**.

If the F₁ hybrids are crossed with **recessive parent**, individuals with **both** the dominant and recessive phenotypes appear in the ratio of 1:1.

While both the crosses are known as Back cross, but the second one is specified as Test cross. The test cross is a cross between heterozygous F₁ hybrid and the double recessive homozygous parent. The test cross is used to determine whether the individuals exhibiting dominant.



DIHYBRID TEST CROSS



A **di-hybrid test cross** (of a heterozygous dominant with pure recessive) gives a phenotypic ratio of **1: 1: 1: 1**

MULTIPLE CHOICE QUESTIONS

- 1) The Dihybrid ratio is represented as
 - a) **9:3:3:1**
 - b) 3:3:9:1
 - c) 9:1:3:3
 - d) 1:9:3:3
- 2) A cross between a hybrid and a pure parent is called _____.
 - a) Monohybrid cross
 - b) Dihybrid cross
 - c) **Back cross**
 - d) Polyhybrid cross
- 3) Test cross is
 - a) A cross between hybrid and dominant parent
 - b) **A cross between hybrid and recessive parent**
 - c) A cross between two hybrids
 - d) A cross between dominant and recessive parent
- 4) Which of the following characteristics of pea plants was not used by Mendel in his experiments?
 - a) seed colour
 - b) seed shape
 - c) **pod length**
 - d) flower position
- 5) The experiment involving two pairs of contrasting characters is

- a) Monohybrid
- b) **Dihybrid**
- c) Back cross
- d) Test cross

6) Back cross to the recessive parents is known as

- a) Crossing over
- b) **Test cross**
- c) Incomplete dominance
- d) Dominance

7) Which one is a test cross

- a) $Tt \times Tt$
- b) $TT \times Tt$
- c) $TT \times TT$
- d) **$Tt \times tt$**

8) Dihybrid test cross ratio is

- a) 3:1
- b) **1:1:1:1**
- c) 3:3:1
- d) 1:1

- 9) In the F₁ plant with the character 'Tt' the plants are tall because
- a) **The dominant gene T masks the expression of the recessive t.**
 - b) The recessive gene t masks the expression of the dominant gene T.
 - c) The dominant T and recessive t genes are expressed simultaneously.
 - d) None of the above
- 10) In a monohybrid cross between tall and dwarf plants, the plants obtained in F₂ generation are represented as
- a) 3 dwarf : 1 tall
 - b) **3 tall: 1 dwarf**
 - c) 3 tall: 3 dwarf
 - d) 3 dwarf : 3 tall
- 11) Across between pure black male and pure white female rats (in which the black colour is dominant) is made. What is the ratio of blacks to white in F₂ generation
- a) All whites
 - b) All blacks
 - c) **3:1**
 - d) 1:3
- 12) Genotypic ratio of monohybrid cross is
- a) 3:1
 - b) **1:2:1**
 - c) 1:1:1
 - d) 9:3:3:1

- 13) Test cross determines
- a) Whether two traits are linked or not
 - b) The genotype of F_2 plant**
 - c) Whether the two species will breed successfully or not
 - d) Number of alleles in a gene
- 14) The Genetic constitution is termed as
- a) Dominant
 - b) Genotype**
 - c) Phenotype
 - d) Allele
- 15) Mendel took _____ contrasting characteristics of pea plants.
- a) eight
 - b) seven**
 - c) six
 - d) five